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Turning Biosolids into Carbon Storage: Scaling Subsurface Injection at the TIRE Facility to Meet Los Angeles' 2050 Population-Driven Emission Demands

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Abstract

The urgency of addressing global climate change, underscored by the 2015 Paris Agreement, coupled with Los Angeles's significant population expansion since 2000 ($\approx 11.17\%$) and anticipated growth through 2050, necessitates the implementation of robust and scalable carbon removal strategies. This study assesses how expanded subsurface Biosolid Slurry Injection (BSI) at the Terminal Island Renewable Energy (TIRE) facility can offset the city's rising Carbon Dioxide (CO_2) footprint. We correlate historical and forecasted population trends with injected volumes and reservoir capacity, presenting a unique city-scale roadmap for transforming Sewage Sludge (i.e., Biosolids) into a durable carbon sink.

To derive a biosolids generation curve, population data spanning 2000 - 2050 were correlated with biosolid production at Los Angeles (LA) County. A life cycle carbon audit was conducted to quantify CO_2 removal (CDR) per barrel injected. To evaluate the maximum injection capacity of the TIRE Facility, reservoir and wellbore simulations were performed across multiple operational scenarios. These included the base case, representing approximately 15% of the facility's current operational capacity, and the full capacity scenario at 100%. The full capacity scenario would be capable of accommodating all current and future regional biosolid sources driven by population growth, thereby supporting regional efforts to achieve emission reduction targets by 2050.

The results demonstrate a linear correlation between Biosolid generation output and regional population growth, with an R^2 value of 0.96. Currently, a single well at the TIRE facility sequesters approximately 0.3 million metric tons of CO_2e annually, corresponding to a Biosolid injection of 150 US tons per day. This setup provides a CDR capacity of $4,684 \text{ Mt CO}_2\text{e yr}^{-1}$, representing only 15% of the Biosolid stream from Los Angeles County. Scaling up pumping capacity to two wells would enable full disposal of LA County's Biosolids, with an injection of 750 US tons daily, sequestering over 1 million $\text{Mt CO}_2\text{e yr}^{-1}$, and achieving a CDR capacity exceeding $30,000 \text{ Mt CO}_2\text{e yr}^{-1}$. Expanding to a seven-well configuration could accommodate regional Biosolid inputs of 7,774 US tons daily, resulting in a total GHG offset exceeding

11 million Mt CO₂e yr⁻¹, and a CDR capacity of approximately 154,442 Mt CO₂e yr⁻¹. These findings underscore the significant potential of the BSI system to meet regional and broader climate mitigation targets through scalable deployment.

The analysis demonstrates the scalability and effectiveness of subsurface BSI technology as a viable strategy for meeting metropolitan-scale CDR targets. The findings confirm that BSI operations can be progressively expanded to align with long-term climate and urban development objectives.

By integrating demographic projections with field proven BSI data, this work presents a unique, city scale roadmap for turning Biosolid into a durable carbon sink. The methodology links municipal Biosolid management and carbon storage engineering, offering a transferable framework for rapidly growing urban areas aiming for net zero emissions through existing infrastructure.

Key Words: Carbon removal, Biosolid slurry injection, TIRE project, Los Angeles Population Growth, Climate and Urban Development, CO₂ footprint audit, Net-zero Emission, Reservoir Simulation

Introduction and Background

Large volumes of municipal and industrial sewage sludge are produced annually on a global scale. Most of this sludge undergoes biological treatment to eliminate dissolved carbon, solids, pathogens, and other contaminants, with the resulting treated effluent discharged into surface water bodies. Effective biological treatment often leads to the incidental formation of nutrient-rich solids, commonly known as Biosolids ([Roy et al., 2011](#)). These Biosolids are generated in substantial amounts as a byproduct of the treatment process. [Gomez et al. \(2010\)](#) reported that the yearly production of dry Biosolids averages between 20 and 32.85 kilograms per person, which requires safe, reliable, and socially acceptable management. According to the United States Environmental Protection Agency (US EPA), Biosolids are processed sewage sludge that meets specific standards for pollutants and pathogens, making them suitable for land application or surface disposal ([US EPA, 2007](#)). As the quality of effluent improves, the volume of Biosolids produced tends to increase, driven by the expansion of sewage sludge treatment facilities and the adoption of more stringent standards for effluent treatment and discharge.

Over recent decades, disposal practices for sewage sludge have undergone significant changes. Prior to 1998, disposal primarily involved discharge into seawaters or use as fertilizer or soil conditioner for agricultural lands, a common practice in Europe ([Odegaard et al., 2002](#)). Alternative methods included incineration and landfilling. The enactment of the European Union's Urban Waste Water Treatment Directive (UWWTD) in 1998 marked a pivotal shift, as it prohibited marine disposal of sewage sludge to protect marine ecosystems. In the United States (US), Biosolids management options include land application, landfilling (with or without energy recovery), composting, anaerobic digestion, heat-dried palletization for fuel or fertilizer, and incineration (with or without energy recovery). In 2004, approximately 80% of Biosolids in the US were managed through land application, landfilling, and incineration ([NEBRA, 2007](#)). Landfilling ensures contained Biosolids management; however, it can result in significant methane emissions if gas capture systems are incomplete. Furthermore, landfills occupy limited space, raising concerns about their long-term environmental and operational sustainability. Land application remains a widely used, cost-effective approach for managing municipal and industrial Biosolids in the US. It benefits the environment by recycling nutrients and carbon into soils and decreasing reliance on landfills. Currently, about 56% of Biosolids are applied for land reclamation, agriculture, and soil enhancement ([US EPA, 2024a](#)). However, operational challenges are increasing. High logistical costs, site restrictions, odors, and vector attraction complicate the practice. Additionally, rising public concern over contaminants such as per- and polyfluoroalkyl substances (PFAS), pharmaceuticals, and metals highlights the urgent need for innovative management strategies to ensure safety and advance environmental sustainability ([US EPA, 2024b](#)).

To address these concerns, regulatory frameworks in the US governing the application of municipal and industrial Biosolids as soil amendments were established to protect individuals who may come into direct contact with Biosolids at land application sites or through the consumption of crops and livestock grown on treated lands. Although controls on Biosolids management have existed since the 1970s, comprehensive regulations regarding the use and disposal of sewage sludge were only enacted in 1993 ([Federal Register, 1993](#)). Since then, these regulations have undergone several revisions, with primary goals including reducing pathogen levels, limiting the appeal of Biosolids to disease vectors such as birds and rodents, and establishing maximum permissible levels for metals and nutrients in Biosolids ([Sain et al., 2025](#); [US EPA, 2024a](#); [Lu et al., 2012](#); [US EPA, 1994](#)). Meanwhile, stricter regulations on landfilling and land application are expected to promote the long-term viability of incineration as a sustainable Biosolids management option ([Malerius and Werther, 2003](#)). Technological advancements have improved process efficiency, reduced plant size, and enhanced energy recovery ([Mininni et al., 1997](#)). However, high capital costs, air pollution concerns, and ash disposal challenges, potentially contributing to environmental degradation, remain significant barriers.

Amid these challenges, climate impact assessments reveal that all surface-based management options contribute to greenhouse gas emissions, notably methane from anaerobic processes and nitrous oxide from nitrogen cycling. Life-cycle assessments show that benefits such as nutrient reuse and energy generation can offset some emissions, underscoring the importance of integrated, sustainable approaches ([Sain et al., 2025](#); [Elgarahy et al., 2024](#); [Roy et al., 2011](#); [Peters and Rowley, 2009](#); [Wang et al., 2008](#)). Table 1, 2, and 3 provide an overview of risk categories and potential mitigation strategies for surface-based Biosolids management options for land application, landfill, and incineration, respectively.

Table 1—Biosolids Management Option: Land Application ([McBride, 2022](#); [Epstein, 2015](#); [Ahring et al., 2002](#); [Valkama et al., 2024](#); [Robinson et al., 2012](#); [Lu et al., 2012](#); [Goldfarb et al., 1998](#); [Harrison and Eaton, 2001](#); [Popoola et al., 2023](#))

Risk Category	Specific Hazard/Description	Primary Controls / Mitigations
Environmental	Nutrient runoff and eutrophication: Excess nitrogen and phosphorus from land-applied Biosolids can leach into water bodies, causing algal blooms and oxygen depletion	Use site-specific management; setbacks; buffer strips; proper timing applications to crop needs & weather conditions; controlled application rates to prevent nutrient runoff & protect water quality
Health & Safety	Pathogen Exposure: Potential infection from bioaerosols or direct contact with harmful microorganisms	Stabilization reduces pathogens and odors via Class A or thermophilic digestion, limiting contact for the public and workers
Climate/Energy	N ₂ O emissions from soils are the release of a potent greenhouse gas produced by microbial nitrogen cycling, contributing to climate change.	Precision application targets nutrients. Nitrification inhibitors slow nitrogen conversion, lowering N ₂ O emissions. Soil moisture management (nitrogen loss)
Social/Community	Odor, pests, and nuisance complaints involve community concerns over smells, pest attraction, and disturbances from Biosolids management.	pH stabilization reduces pathogens and odors. Timing with weather reduces risks. Community notice keeps residents informed. Vector control prevents pests.
Regulatory/Legal	Non-compliance involves exceeding application limits, buffer zones, or planting restricted crops	Regular maintenance, trained personnel, strategic siting, precise documentation & performance reviews
Technical/Operational	Emerging contaminants (PFAS, microplastics, metals) in Biosolids can accumulate in soil, affecting soil health and entering the food chain.	Upstream control reduces pollution; standards limit pollutants; long-term soil testing monitors contamination; crop selection; avoid vulnerable sites

Table 2—Biosolids Mangmanet Option: Landfill (Walsh et al., 2020; Ogle et al., 2025; Moqbel, 2009; Cai et al., 2018; Majumder et al., 2014; Basavaraju et al., 2025; Epstein, 2015; Ekinci and Arslankaya, 2022)

Risk Category	Specific Hazard Description	Primary Controls / Mitigations
Environmental	Leachate with PFAS/metals is contaminated liquid seeping from soils or Biosolids sites, complicating treatment and risking environmental contamination.	Composite liners prevent leachate leaks; collection systems gather & treat leachate; pre-acceptance testing ensures Biosolids compliance; pre-treatment removes contaminants like PFAS before discharge.
Health & Safety	Landfill fires/self-heating in mono-fills risk spontaneous combustion from heat generated by anaerobic decomposition of Biosolids.	Biosolids acceptance controls ensure safe disposal; temperature/CO monitoring detects fire risks; spacing and compaction reduce oxygen and heat build-up
Climate/Energy	HC ₄ is produced during anaerobic Biosolids decomposition; fugitive emissions are unintended HC ₄ releases from leaks	Efficient landfill gas (LFG) capture; oxidation covers oxidize residual HC ₄ ; energy recovery converts captured HC ₄ into electricity or heat
Social/Community	Odor, vectors, birds, and traffic are concerns related to Biosolids management; unpleasant smells, pest attraction, wildlife disruption, increased transportation traffic.	Daily cover prevents odors and pests; misting controls dust and bioaerosols; traffic routing minimizes community impact; community engagement involves informing and involving residents.
Regulatory/Legal	Long-term liability involves monitoring and managing contamination and gas emissions to ensure environmental safety after site closure	Financial assurance funds site closure and liabilities; conservative design ensures safety margins; robust documentation maintains transparency and compliance.
Technical/Operational	High-moisture Biosolids can cause land subsidence and slope instability, risking structural failure and environmental contamination	Limit wet tonnage sets max Biosolids volume for safety; dewater Biosolids for handling; assess slope stability; use phased filling to prevent failure.

Table 3—Biosolids Mangmanet Option: Incineration (Kamaruddin et al., 2023; Roy et al., 2011; Fernandez-Anez et al., 2020; Hoener, 2024; Mohamed et al., 2024; Gillett, 1992; Pozzebon and Seifert, 2023; Liu et al., 2022)

Risk Category	Specific Hazard Description	Primary Controls / Mitigations
Environmental	Air emissions include particulate matter (PM), NO ₂ , SO ₂ , HCl, toxic metals, and dioxins and furans, posing health and environmental risks	Advanced Air Pollution Control (APC), such as fabric filters (baghouses); Real-time Emissions Monitoring Systems (EMSs) track pollutant levels for compliance.
Health & Safety	Explosion or flash events occur when flammable vapours from volatile Biosolids ignite during high-temperature operations	Feedstock screening ensures Biosolids meet safety standards; Lower Explosive Limit (LEL) monitoring detects combustible gases to prevent explosions.
Climate/Energy	GHG (i.e., CO ₂ , CH ₄ , and N ₂ O) emissions	Optimize Combined Heat and Power (CHP) systems for efficiency; Moisture reduction enhances stability
Social/Community	Community resistance arises from health, safety, environmental concerns, like fears of pollution and odors	Transparent portals enable real-time monitoring; community benefits include jobs and pollution reduction; independent (unbiased) audits
Regulatory/Legal	Permit limit exceedances and unforeseen failures can temporarily raise emissions or hazards	Backup APC systems ensure continuous emission control; Developing procedures to effectively handle operational disruptions
Technical/Operational	Ash management involves safely handling residues with metals and PFAS to prevent environmental contamination and protect health.	Stabilization/solidification immobilizes hazardous substances like metals and PFAS, reducing leaching; PFAS-optimized APC technologies enhance control

In this context, deep subsurface Slurry Injection (SI) of Biosolids (i.e., Biosolid Slurry Injection or BSI) has emerged as an innovative alternative to conventional surface methods. The Los Angeles Terminal Island Renewable Energy (TIRE) project exemplifies this pioneering approach. In this initiative, treated Biosolids are injected into a deep, saline formation approximately 5,300 feet below the surface. This method isolates Biosolids from potable groundwater and surface ecosystems, allowing natural geothermal heat and pressure to facilitate thermophilic anaerobic biodegradation within a confined geological setting. This innovative strategy offers a potentially safe, sustainable, and low-risk pathway for Biosolids management, addressing many limitations of traditional surface-based practices.

The TIRE Project: Achieving 2050 Climate Goals with Urban-Scale Negative Emission Technology (NET)

The urgency of addressing global climate change, underscored by the 2015 Paris Agreement (United Nations, 2015), coupled with Los Angeles's significant population expansion since 2000 ($\approx 11.17\%$) and anticipated growth through 2050, necessitates the implementation of robust and scalable carbon removal strategies. Los Angeles (LA), LA County, and California have historically experienced unprecedented and sustained population growth, a phenomenon that is unmatched among similarly developed regions globally over an extended period. This growth has significant implications for regional infrastructure, resource management, and environmental sustainability. According to projections from the California Department of Finance (CA DOF), the state's population is anticipated to continue its upward trajectory over the next two decades, reaching approximately 42 million by 2050 (CA DOF, 2025).

Figures 1a and 1b illustrate the population trends of LA City, LA County (highlighted in red), and the state of California across the periods 2010–2025 and 2000–2050, respectively (CA DOF, 2025). It is worth noting that the light orange region in Figure 1a represents areas with population growth of less than 10 (e.g., LA County), while the dark orange regions experienced population increases equal to or exceeding 10%. Figure 1b indicates that the estimated populations in 2025 are approximately 5 million for LA City, 10 million for LA County, and 36 million for California. Projections for 2050 suggest these figures will increase to approximately 6 million, 12 million, and 42 million, respectively. Despite a projected decline in the annual growth rate to approximately 1%, the total increase remains significant, with an annual growth of over 350,000 residents on average ($\pm 10\%$) in California (Johnson, 2003; PPIC, 2015).

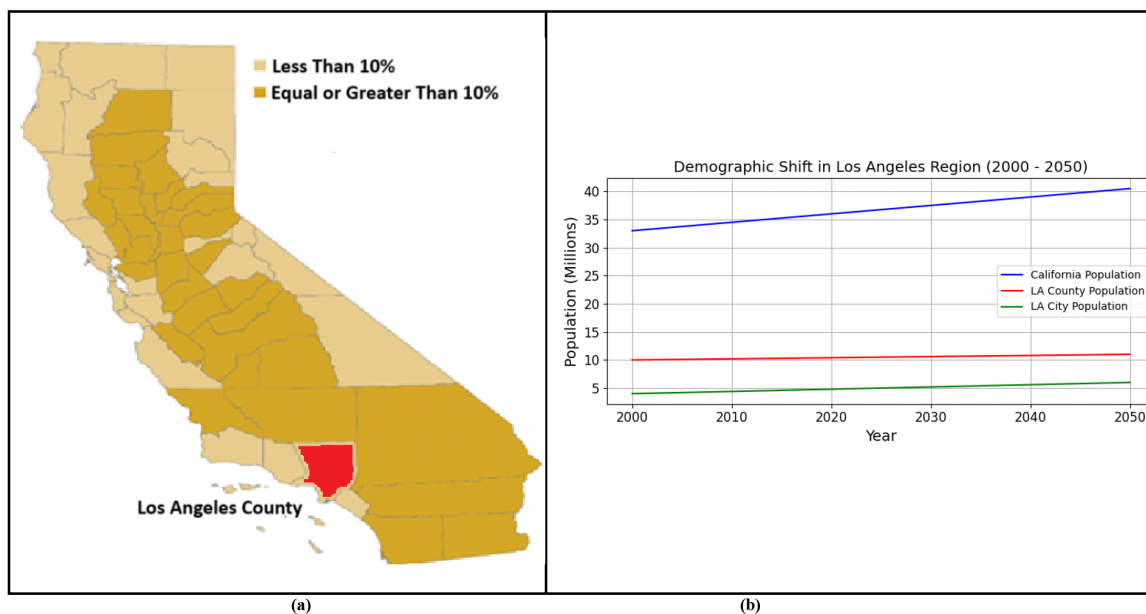


Figure 1—Population Projections and Future Trends: (a) Demographic projections trend of California from 2010 to 2025 (b) Demographic Shift in California, Los Angeles County, and Los Angeles City from 2000 to 2050 (CA DOF, 2025; PPIC, 2015)

Simultaneously, as urban populations continue to expand and environmental concerns intensify, the need for innovative and sustainable Biosolids management solutions has become increasingly urgent (Struglia & Winter, 2002). The TIRE project in LA City exemplifies innovative approaches to Biosolids management and renewable energy production by converting Biosolids into a valuable resource via deep subsurface Biosolid Slurry Injection (BSI). This process is further supported by the US Environmental Protection Agency (EPA) and Los Angeles City's Underground Injection Control (UIC) program, which ensures regulatory compliance and provides long-term containment of injected materials (EPA, 2020; City of Los

Angeles UIC, 2013); Biosolids, which are highly treated organic residuals derived from sewage sludge treatment processes, are being combined with treated, non-hazardous municipal and industrial sewage sludge residual solids and effluent to form slurry mixtures (Panchal, 2021; Brown et al., 2020; Panchal et al., 2018; Fitzmorris et al., 2009). These are injected into geological formations at LA County, operated by the TIRE project. At a depth of approximately a mile, the Biosolids undergo a natural, high-temperature anaerobic biodegradation process within the saline environment of the deep formation. This transformation yields methane (CH_4) and carbon dioxide (CO_2) from organic matter, alongside stable residual solids. The CO_2 is permanently captured via dissolution into the formation's saline water. Concurrently, the comparatively pure CH_4 migrates and becomes securely trapped within the reservoir, potentially serving as an environmentally safe option. This approach not only provides a practical solution for the disposal of Biosolids but also harnesses subsurface conditions to produce renewable energy and sequester atmospheric CO_2 . By addressing both Biosolids management challenges and climate mitigation goals, the TIRE initiative plays a vital role in advancing sustainable urban infrastructure and supporting LA County's broader climate action objectives.

To achieve the ambitious climate targets of limiting global warming to 1.5°C - 2°C as outlined in the Paris Agreement (Klein et al., 2017), rapid and substantial reductions in greenhouse gas (GHG) emissions are essential (IPCC, 2021; Keith et al., 2018; Rogelj et al., 2018). This necessitates a comprehensive, multi-pronged approach that includes transitioning to renewable energy, enhancing energy efficiency, and transforming land use practices. Amid these efforts, the development and deployment of large-scale Negative Emissions Technologies (NETs) are increasingly critical (Snyder, 2022; Haszeldine et al., 2018). These technologies must be capable of sequestering billions of tonnes of CO_2 annually to offset residual emissions and ensure climate goals are met (Hilaire et al., 2019; Rogelj et al., 2018). Currently available NET options, such as direct air capture (DAC), bioenergy with carbon capture and storage (BE-CCS), afforestation, soil carbon management, and mineral weathering, each face significant limitations (Fuss et al., 2018). For instance, DAC is energy-intensive due to the low ambient CO_2 concentration (~ 410 ppm), which constrains its scalability (House et al., 2011). The BE-CCS relies heavily on biomass availability and requires substantial energy for CO_2 separation and compression. Land-based approaches like afforestation and soil management are limited by land availability and are vulnerable to carbon loss, especially in aboveground stocks (Snyder, 2022; Hepburn et al., 2019). Furthermore, the economic feasibility of CO_2 utilization for Enhanced Oil Recovery (EOR) in the Oil and Gas (O&G) industry largely depends on its ability to enhance hydrocarbon extraction and the market value of produced resources. In certain US fields, naturally occurring CO_2 from gas reservoirs is considered a cost-effective and preferred source over anthropogenic options. However, capturing CO_2 from DAC and industrial processes, such as steel, cement production, refineries, and power generation, is a mature and proven technology that supports large-scale Carbon Capture, Utilization, and Storage (CCUS) initiatives. These approaches are critical for emissions reduction, resource recovery, and long-term geologic sequestration, leading to the concept of "green oil," where underground storage surpasses emissions (Andersen et al., 2020). Despite technological progress, significant challenges limit widespread adoption. High capital costs, long-term liability, stringent monitoring and permanence requirements, along with public concerns over induced seismicity and potential leakage, pose substantial barriers to scaling up CO_2 -based EOR and CCUS implementation.

In this landscape of constraints, BSI has emerged as a promising and potentially indispensable NET. It offers an efficient means of sequestering large volumes of CO_2 within stable, long-term geological formations, bypassing many land use and energy limitations associated with existing methods. Figure 2 shows the incremental cost ($\$/\text{tCO}_2$) analysis relative to annual carbon storage capacity, measured in gigatons (Gt), for a selection of currently available NETs (Snyder, 2022; Hepburn et al., 2019; Fuss et al., 2018). As global demand for scalable and cost-effective negative emissions solutions intensifies, innovative approach such BSI is gaining increasing recognition for their potential to contribute significantly to global

climate mitigation efforts. This approach not only offers a practical solution for biosolid disposal but also leverages subsurface conditions to generate renewable energy and sequester atmospheric CO₂. By addressing Biosolids management and climate mitigation simultaneously, the TIRE initiative advances sustainable urban infrastructure while supporting Los Angeles's broader climate action goals.

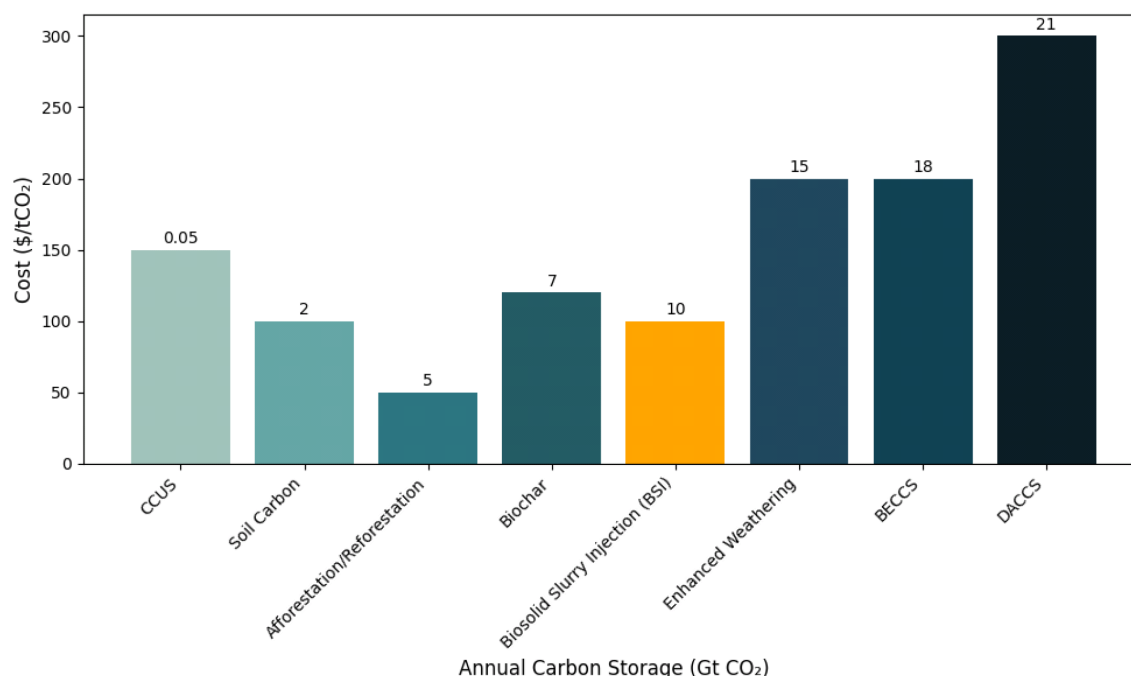


Figure 2—Cost Analysis of Negative Emissions Technologies, NETs
(Reproduced from [Snyder, 2022](#); [Hepburn et al., 2019](#); [Fuess et al., 2018](#))

The objective of this study is to evaluate the feasibility and scalability of deep subsurface BSI as an urban-scale NET for carbon removal (i.e., maximize GHG offsets). By correlating population growth from 2000 to 2050 with biosolid production at LA County, we develop a Biosolids generation curve and perform a comprehensive life cycle carbon audit to quantify CO₂ sequestration per unit of injected material. Utilizing reservoir and wellbore simulations, multiple-stage operational scenarios including a baseline case, representing approximately 15% of the facility's current capacity, and a full-capacity case at 100%. The full-capacity scenario is designed to accommodate both existing and future Biosolids volumes driven by regional population growth. This expanded capacity would support regional efforts to achieve significant emission reductions and meet climate targets by 2050.

The analysis indicates that BSI operations at the full capacity of the TIRE project can effectively contribute to large-scale carbon removal goals while emphasizing the importance of monitoring for containment and efficiency ([Abou-Sayed et al., 2002a](#)). Ultimately, this work presents a city-scale, adaptable framework that links municipal Biosolids management with carbon storage, supporting urban efforts to achieve net-zero emissions through existing infrastructure.

Biosolids Slurry Injection (BSI) Technology

According to the Environmental Protection Agency (EPA) O&G production in the US involves the disposal of approximately 2 billion gallons of related sewage sludge daily via underground injection wells ([EPA, 2020](#)). However, the injection of slurry mixtures presents distinct technical challenges not encountered in conventional sewage sludge disposal, primarily the risk of solid particles clogging pore spaces and damaging the geological formations ([Xia, 2007](#); [Abou-Sayed et al., 2002b](#); [Abou-Sayed and Guo, 2001](#)). To address this issue, Biosolids Slurry Injection (BSI) technology has been developed. This process begins by injecting

clean water (pre-flush) to condition the formation prior to receiving the Biosolids. Subsequently, a slurry containing Biosolids is injected, with the Biosolids particles becoming trapped within the fractures.

To remove potential Biosolids build-up in the wellbore, a final flush (post-flush) with clean water is performed at the end of each injection cycle. This ensures that all Biosolids are effectively trapped within the formation. This cycle consists of approximately 10-12 hours of active injection, followed by a short flush, then a 12-hour shut-in period. During the shut-in, the injected water gradually leaks off, reducing the pressure below the fracture threshold and facilitating the sealing of the fractures (Mohamed et al., 2018). This cycle is repeated daily. Unlike hydraulic fracturing in shale gas extraction, which relies on higher pressures to create extensive, long-reaching fractures, BSI employs lower pressures to produce shorter, more controllable fracture networks near the wellbore. Through successive cycles, these fractures can gradually propagate, offering an effective method for storing significant volumes of solids within the subsurface (Kordzi & Region, 1998).

Selecting an appropriate formation is critical for the success of slurry injection operations, with hydraulic isolation from underground sources of drinking water (USDWs) being essential to prevent contamination. Suitable formations should exhibit low permeability and possess favourable geomechanical properties, such as high Young's modulus, Poisson's ratio, and in-situ stress. A site-specific geomechanical model assesses in-situ stress conditions and predicts fracture initiation pressures, which are exceeded when injection pressures surpass formation stress thresholds. The Step Rate Test (SRT) further aids this process by plotting wellhead injection pressure (WHIP) against injection rate, identifying the pressure point at which fractures may initiate, thereby ensuring operation within safe limits and maintaining formation integrity.

Since at least 1989, SI has been employed to dispose of various geogenic slurries (Marinello et al., 2001), including drilling fluids, cuttings, naturally occurring radioactive materials, and contaminated soils (Baker et al., 1999; Bruno et al., 1995; Dusseault, 1998; Gaurina-Med et al., 2020; Reed et al., 2002; Sipple-Srinivasan et al., 1998; Veil & Dusseault, 2003). Typically, field applications are short-term, confined to drilling operations. However, there are cases of long-term use; notably, ConocoPhillips operated SI systems in Alaska from the 1980s through 2003, with one well injecting over 6.2 million barrels of slurry materials (~ 1 million m_3) at approximately 3,500 barrels per day (Veil & Dusseault, 2003). Moreover, a study by Bruno (2002a), analysing data from eight injection wells and over 500 injections, found little evidence of increased formation pressure or declining injectivity over extended periods, indicating that long-term injection is feasible (Bruno et al., 2002b).

Most relevantly, since July 2008, the TIRE project has employed BSI as a method to dispose of biogenic sewage slurries. Injection occurs through an injection well extending approximately a mile below the surface, with an injection capacity of up to 600 US-ton per day and a biosolid concentration between 5% and 10% by weight (Panchal et al., 2018). From 2008 to date, roughly 972,300.00 US-ton of Biosolids were injected via a single well (Los Angeles UIC, 2018). Three additional wells function as backups and monitoring wells. To date, there has been no observed correlation between injection activities and an increase in seismic events. The TIRE project's successful long-term operation demonstrates that deep biosolid injection is a technically viable and effective method for Biosolids disposal and carbon sequestration when conducted in suitable geological formations (Panchal et al., 2021; Panchal et al., 2020a).

Biosolid Management System and Injection Workflow Scenarios

The proposed approach involves collecting Biosolids from LA City's production or other processing facilities. This Biosolids material is then transported to TIRE's injection site. Upon arrival at the site, additional treatment (pre-treatment) of the Biosolids may be required, depending on the specific feedstock. Subsequently, the biosolid is combined with water to create a Bioslurry suitable for injection; Ideally, the water used would come from sewage sludge effluent produced by municipal treatment plants, which would otherwise be discharged into surface waters. Alternatively, seawater or excess surface water can be used.

The resulting Bioslurry is injected into carefully selected underground formations. At typical temperatures (below $\sim 75^{\circ}\text{C}$), anaerobic biodegradation converts organic solids into CH_4 , CO_2 , and digestate, which may influence the formation's storage capacity and sealing efficiency. While higher temperatures would promote thermal decomposition in deeper formations. Formation selection, guided by geomechanical and fracture simulation models, ensures safe and effective injection. As demonstrated at the California-based TIRE facility, this carbon dioxide equivalent (CO_2e) sequestration, via existing slurry, is environmentally sustainable and economically viable through tradable carbon offsets. By converting organic solids into sequestered CO_2 and CH_4 , this process significantly reduces atmospheric GHGs, supporting carbon markets and climate mitigation efforts.

This study evaluates five distinct operational scenarios for a BSI process, visualized in the schematic workflow diagram, as shown in Figure 3. Scenario 1 reflects the baseline operation of the TIRE project, with an average daily injection of 150 US tons of dry Biosolids. This scenario provides a reference framework for assessing the potential scalability and capacity expansion of the BSI process. Subsequent scenarios explore TIRE's capacity to accommodate biosolid material from both local and regional sources within California, through a sensitivity analysis. Scenario 2 increases the daily injection to 600 US tons. Building on this, scenario 3 considers the total Biosolids produced across LA County, with a projected daily injection of 750 US tons. Scenario 4 models an increase in daily injection capacity, doubling from 750 to approximately 1,500 US tons, achieved by utilizing two wells to accommodate the higher Biosolids volume. Finally, scenario 5 explores full utilization of TIRE's maximum capacity, with an injection rate of 7,774 US tons per day across seven wells. These scenarios facilitate a comprehensive understanding of the system's scalability under varying operational demands.

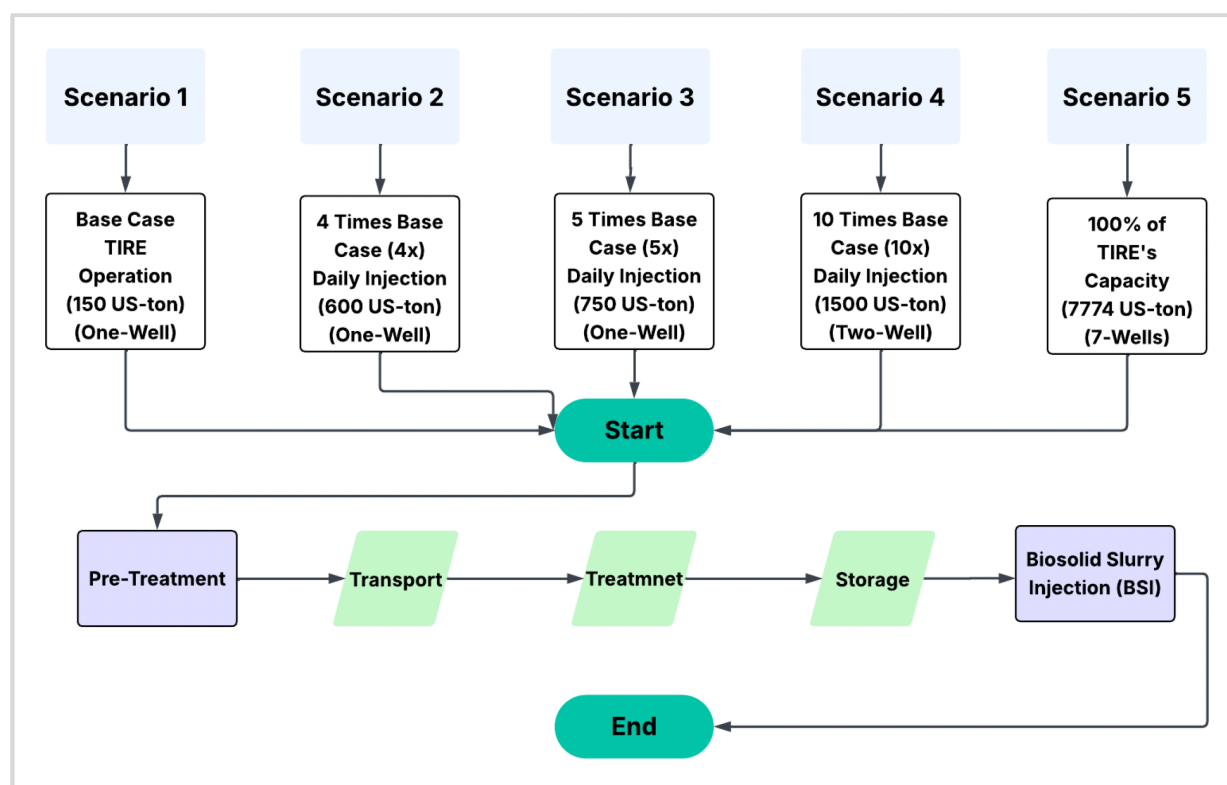


Figure 3—Schematic of the Five-Scenario Model Used to Calculate GHG Reductions and Carbon Credits for the TIRE Project in Los Angeles, California

It's worth noting that a single well in the TIRE Facility, representing 14.28% of the facility's total capacity, can process approximately 2,400,000 US tons of Biosolids, as shown in Figure 4. Therefore, the current

operation, injecting 150 US tons per day, utilizes only 14.5% of the total capacity of a single well and just 1.75% of the entire TIRE facility's seven-well capacity. Model overview across the TIRE's operation scenarios is summarized in Table 4. Furthermore, for each scenario modelled, we collected operational data from the TIRE facility spanning 2022 through 2024. We also assume that the total biosolid material delivered to TIRE and other regional facilities represents the total Biosolids collected within the specified region, as detailed in Table 5.

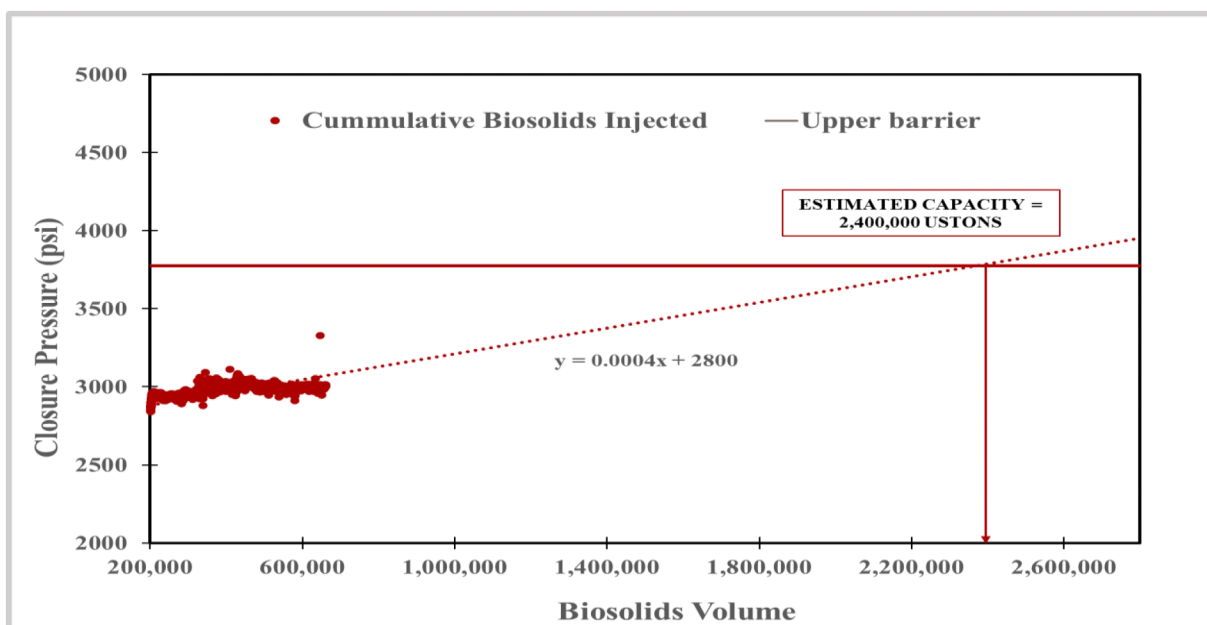


Figure 4—Relation Between Closure Pressure (psi) and Biosolids Volume (US-ton) for Estimation Well injection Capacity

Table 4—TIRE's Operational Scenarios: Model Overview

No.	TIRE's Operational Scenario				
	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5
TIRE's Daily Capacity (US-ton)	150	600	750	1,500	7,774
Operational Days	2,161	2,161	2,161	2,161	2,161
Total Capacity per Operational Days (US-ton)	324,150	1,296,600	1,620,750	3,241,500	16,799,614
Capacity Usage (%) (1 – Well)	14.5	54	67.5	135 (more than on well capacity)	-
Capacity Usage (%) (7 - Wells)	1.75	7.7	9.6	19.2	100

Table 5—Total Biosolids Received at TIRE& Other Facilities (LA City, California, US)

Year	Biosolids Received at TIRE (US-tons)	Biosolids Received at Other Facilities (US-tons)	Total Biosolids (US-tons)	Biosolids Received at TIRE (%)
2022	48,506	227,442	276,948	17.5
2023	53,289	196,271	249,560	21.36
2024	49,209	217,530	266,739	18.45

All scenarios progress through a standardized sequence biosolid material of pre-treatment, transport, treatment, storage, and BSI (Figure 3.) These workflow scenarios emphasize the scalable nature of BSI, from current operations to levels designed to maximize throughput and resource utilization. Notably, the

scalability demonstrated by these scenarios suggests that TIRE possesses the potential to manage biosolid material from Los Angeles City and County, the wider California region, and potentially beyond, even when considering population growth projections through 2050.

Technoeconomic Analysis

To evaluate the economic viability of a Biosolids-based sustainable BSI system, following [Snyder \(2022\)](#), we calculated its net present value (NPV) over 30 years using a spreadsheet model. The system comprises four main modules: Pre-treatment (i.e., depending on the specific feedstock), transportation, treatment & Storage (i.e., Biosolids are mixed with water to produce a Bioslurry suitable for injection and stored in tanks prior to being transferred to the injection site), and injection, aligned with regional Biosolids production in the US. Key assumptions include:

- Feedstock input: Daily delivery of 150 US tons of digested sludge and wet cake from LA County to TIRE's facility
- Transportation: \$0.20 per mile per ton over an average distance of 25 miles ([US DOT, 2021](#)).
- Treatment & Storage: Water is assumed to be supplied to the BSI system at no cost enabling suitable Bioslurry production.
- Drilling and injection: Drilling costs for US wells were estimated using the formula provided by the American Petroleum Institute (API), with a typical cost of \$1,232 per foot for 5,000 ft wells. Injection wells have a 5-year lifespan, with capacity of 312,500 tonnes of solids per well ([API, 2008](#)).
 - $\text{Cost}_{\text{well}} = (0.00024 * \text{Depth}^2 + 0.0528 * \text{Depth} - 131.45) * 100 = 6,013,255 \text{ \$ per well (well}^{-1}\text{)}$ ([API, 2008](#))
- Operational costs: include pump operation, well maintenance assumed at 10% of initial well costs, and well workovers included ([Snyder, 2022](#)).
- Regulatory and permits: Costs for well plugging and abandonment incorporated; long-term monitoring costs are excluded.

The input data along with the essential model parameters and assumption are summarised in [Table 6](#).

Table 6—Summarized input data applied in the Technoeconomic Analysis

Parameter	Description		
	Value	Unit	Source
Total Biosolids	150	US tons	TIRE's Project
Truck Transport Cost	0.2	\$ ton ⁻¹ mile ⁻¹	US DOT, 2021
Biogas Content (CO ₂ & CH ₄)	65 & 35	%	TIRE's Project
Injection Rate	8.4	bbl min ⁻¹	Panchal et al., 2020a
Biosolid Slurry Density	75.0	lb/ft ³	TIRE's Project
Well Depth	5400	ft	Panchal et al., 2020b
Drilling & Completion CAPEX	6,013,255	\$ well ⁻¹	Calculated Follow API, 2008
Sewage Sludge pump & Pipeline CAPEX	1,056,000	\$	BBC Research, 2016
Insurance, pore lease, and additional OPEX	300,000	\$ yr ⁻¹	Snyder, 2022
Other Maintenance	10	10% of CAPEX	Snyder, 2022
Decommissioning Cost	30,000	\$ well ⁻¹	Raimi, 2021
Biosolids-handling	100,000	\$ per 150,000 ton	Assumed

Parameter	Description		
	Value	Unit	Source
Pump Size	1000	hp well ⁻¹	Panchal et al., 2020a
Pump Operational Time	12	Hours day ⁻¹	Panchal et al., 2020b
Electricity price	0.14	\$kWh ⁻¹	DOE, Electricity Data, 2021
Labour (6 employees)	60,000	\$ yr ⁻¹	Assumed
Carbon Assurance	1	\$ ton ⁻¹	Snyder, 2022
Debt / Equity	50/50	-	Assumed
Loan Term	10	yr	Snyder, 2022
Tax and Discount Rate	22 & 10	%	Damodaran, 2025

Based on the specified assumptions and assuming revenue is generated per net metric ton (MT) of CO₂ stored, the technoeconomic analysis indicates that the BSI process has a breakeven CO₂ price of \$55.00 per US ton. It can inject an average of 50,000 US tons of dray Biosolids annually, corresponding to approximately 341,329 MT of CO₂ equivalent offsets over 20 years (i.e., GHG Offset in MT CO₂e-20 Year), utilizing a single injection well. Furthermore, the BSI model produced cost estimates ranging from \$38 to \$96 per ton of CO₂ equivalent (CO₂e), with variability driven primarily by feedstock biosolid content. Overall, the process demonstrates economic viability within this cost range. These results are shown in Figure 5a and 5b. The technoeconomic analysis identifies injection capacity, well lifespan, and feedstock Biosolids content, as the primary factors influencing overall costs. These variables significantly affect project feasibility and economic performance, highlighting the importance of optimizing operational parameters to enhance cost-efficiency.

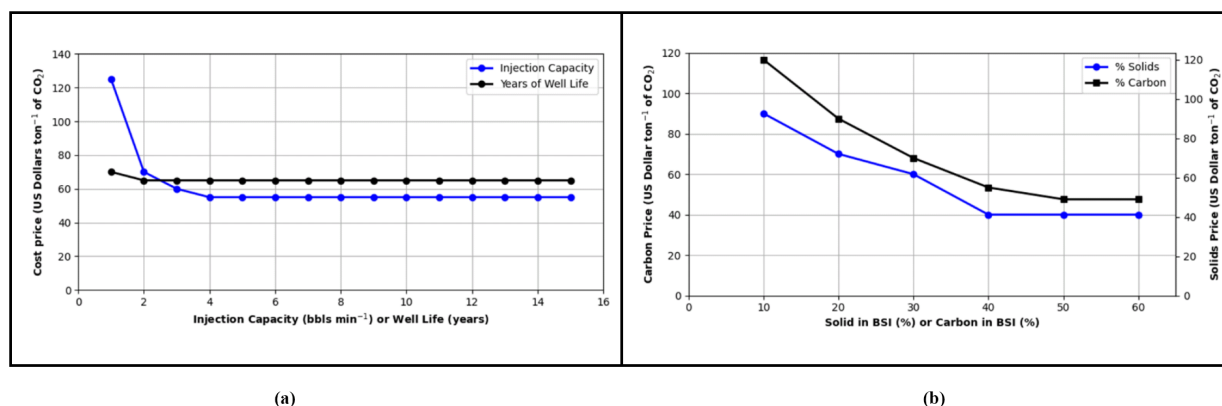


Figure 5—Impact of Injection Capacity, Well Life, Solids Content, and Feedstock Carbon on Technoeconomic Performance (Reproduced from Snyder, 2022; Hepburn et al., 2019; Fuess et al., 2018)

Life Cycle Emission of Biosolid Slurry Injection (BSI)

The objective of a life cycle emissions assessment (LCA) is to comprehensively evaluate potential greenhouse gas (GHG) generation, mainly CO₂ and CH₄, released throughout the entire life cycle of a BSI process, from raw material extraction to sequestration. The full lifespan of the BSI process includes Biosolids collection and generation, treatment, transportation (where applicable), injection, and long-term storage. This assessment aims to provide a holistic understanding of the overall environmental footprint associated with the complete BSI system.

This LCA highlights the environmental and economic benefits of the TIRE facility, including cost savings compared to conventional surface and subsurface disposal methods. Additionally, the system's capability

to safely store CO₂ and CH₄ underground prevents their release into the atmosphere over the long term. By reducing transportation requirements and land application needs, it further minimizes CO₂ and CH₄ emissions. The land application offset provides an additional environmental benefit, supporting negative emissions efforts. Collectively, these advantages contribute to the TIRE process's role in achieving the 2050 emission reduction targets.

By quantifying GHG emission reductions and evaluating economic benefits, the LCA informs environmentally conscious decisions and demonstrates the TIRE facility's contribution toward sustainability goals.

Furthermore, the TIRE Facility processes sewage sludge from multiple facilities within LA County, handling a blend of wet and hydrated Biosolids quantified in US tons. This biosolid input produces biogas composed of approximately 65% CO₂ and 35% CH₄ by volume, both of which are potent greenhouse gases (GHGs) that significantly impact environmental sustainability.

To quantify the overall environmental savings of the BSI process, the model accounts for these primary GHG emission sources, focusing on CO₂ and CH₄, while using the global warming potential (GWP) index which assesses the heat-trapping capacity of a gas relative to CO₂ over a defined period, and that varies based on selected time horizons. Therefore, by summing individual offsets achieved through biogas production, reduced land application, and decreased transportation needs, we get the total GHG offset.

To calculate the total GHG offset (i.e., CO₂ and CH₄) in metric tons of CO₂-equivalent (CO₂e) for a given biosolid injection volume, considering direct emissions from biogas, land application, and transport, the following steps were employed:

1. Convert Biosolids from US tons to Metric Tons (MTs).
2. Calculate Total Biogas Production Volume in Standard Cubic Foot (SCF).
3. Determine CO₂ and CH₄ Volume in SCF.
4. Convert CO₂ and CH₄ Volume from SCF to MT
5. Calculate GHG offset in MT of CO₂e.
6. Calculate Land application Offset in MT of CO₂e
7. Calculate Transport-related GHG Offset in MT of CO₂e

The final total GHG offset is calculated as the sum of GHG generation from biogas, land application offset, and transport savings, following two scenarios:

- Total avoided GHG at TIRE (MT CO₂e - 20-year GWP = Avoided CO₂ by Injection (MT CO₂e - 20-year GWP) + Offset for Land Application (MT CO₂e) + Total avoided CO₂e from transport (MT CO₂e - 20-year GWP).
- Total avoided GHG at TIRE (MT CO₂e - 100-year GWP) = Avoided CO₂ by Injection (MT CO₂e - 100-year GWP) + Offset for Land Application (MT CO₂e) + Total avoided CO₂e from transport (MT CO₂e - 100-year GWP).

To calculate Carbon Dioxide Removal (CDR) credits generated by TIRE is as follows:

- Total avoided carbon at TIRE = Total Biosolids * Biosolids percent solids * Percent volatile solids * Carbon Coefficient / Conversion factor from US Ton to Metric Ton (MT)

The input data along with the essential model (i.e., scenario 1) parameters and assumption are summarised in [Table 7](#).

Table 7—Summarized Input Data for Total GHG Offset and Carbon Credit Calculation (Scenario 1)

Parameter	Description		
	Value	Unit	Definition
Biosolid	1	US-ton	The unit measure of Biosolids in US-ton, typically treated organic material from sewage sludge
Biosolids percent solids	28	%	The percentage of dry solid in the Biosolids, indicating its concentration and water content
Percent volatile solids	62	%	Percentage of solids in Biosolids that can be vaporized or decomposed by microorganisms, producing CO ₂ and CH ₄
Biogas per ton of volatile solids	34,800	SCF/US-ton	Measures the volume of biogas produced from the anaerobic digestion of one US-ton of organic, degradable solids in Biosolids
Biogas Percent CO ₂	35	%	Percentage of CO ₂ present in the biogas mixture
Biogas Percent CH ₄	65	%	Percentage of CH ₄ present in the biogas mixture
Percent volatile solids destroyed	100	%	Measures the proportion of organic solids decomposed during treatment, indicating the efficiency of biogas production
CO ₂ /ton Biosolids injected	2,114	SCF/US-ton	Measures the amount of CO ₂ sequestered per US-ton of Biosolids injected, indicating the system's CO ₂ removal (CDR) efficiency (SCF/US-ton)
CO ₂ /ton Biosolids injected	0.11	MT/US-ton	Measures the amount of CO ₂ sequestered per US-ton of Biosolids injected, indicating the system's CO ₂ removal (CDR) efficiency (in MT/US-ton)
CH ₄ /ton Biosolids injected	3,927	SCF/US-ton	Measures the amount of methane produced per US-ton of Biosolids processed during anaerobic digestion (in SCF/US-ton)
CH ₄ /ton Biosolids injected	0.08	MT/US-ton	Measures the amount of methane produced per US-ton of Biosolids processed during anaerobic digestion (in MT/US-ton)
CO ₂ e factor for methane < 20 years	84	MT CO ₂ e	The short-term Global Warming Potential (GWP) of methane, quantifying its climate impact relative to CO ₂ over a period of less than 20 years (in MT CO ₂ e)
CO ₂ e factor for methane - 100 years	28	MT CO ₂ e	Measures methane's long-term warming effect relative to CO ₂ over a 100-year period (in MT CO ₂ e)
CO ₂ e factor for N ₂ O - 20 to 100 years	298	MT CO ₂ e	Measures N ₂ O's warming potential relative to CO ₂ over a 20-to 100-year period (in MT CO ₂ e)
Tons carried in a truck	25	US-ton	The maximum weight of cargo a truck can transport per trip
Tons carried in 6 trucks per day	150	US-ton	Indicates the total cargo weight transported daily by six trucks
Road miles from Biosolids Source to alternate facility (one way)	119	miles	The distance travelled in miles to transport Biosolids from the source to an alternative site.
Fuel efficiency (diesel truck)	7.5	mpgal	Measures how far a truck travels per unit of diesel fuel, typically in mile per gallon (mpg) or L/100 km
CO ₂ Emissions per gallon of diesel	10.2	kg/gal	Quantify the amount of CO ₂ released when one gallon of diesel is burned (in kg/gal)
Offset for Land Application	0.07	MT CO ₂ e/MT injection solid	Measures the CO ₂ equivalent reductions per metric ton of Biosolids applied to land, indicating environmental benefits over other disposal methods
Conversion Factor from US-ton to MT	1.102	-	Converts short tons or US-ton to metric tons (MT), with 1 US ton roughly equal to 0.9072 MT
Mobile Combustion CH ₄ - diesel Medium and Heavy-duty Trucks	0.0095	g/mile	The methane (CH ₄) emissions from diesel trucks during operation, contributing to GHG emissions
Mobile Combustion N ₂ O - diesel Medium and Heavy Duty-Trucks	0.0431	g/mile	The N ₂ O emissions produced by diesel trucks during operation, contributing to GHGs

Model Results: Carbon Dioxide (CO₂) and Methan (CH₄) Emissions across BSI Scenarios

Figures 6 present the model outputs for CO₂ and CH₄ generation under TIRE's baseline operational scenario (scenario 1), spanning the years 2022-2024. Moreover, Figure 7 compares the total biosolid injection (Figure 7a) and corresponding CO₂ and CH₄ generation (Figure 7b) across five different scenarios, illustrating the impact of varying operational capacities. As expected, CO₂ and CH₄ production is largely proportional to the injection volume. This trend holds true across all scenarios. The scenario-based approach allows for evaluating the potential impact of scaling up BSI operations. These high-volume scenarios (scenario 2, 3, 4, and 5) demonstrates the TIRE facility's potential to reduce GHG in the atmosphere and the potential to mitigate environmental challenges. Given its scale, scenario 5 predicts substantial greenhouse gas reduction, potentially yielding a significant positive impact on the environment and public health.

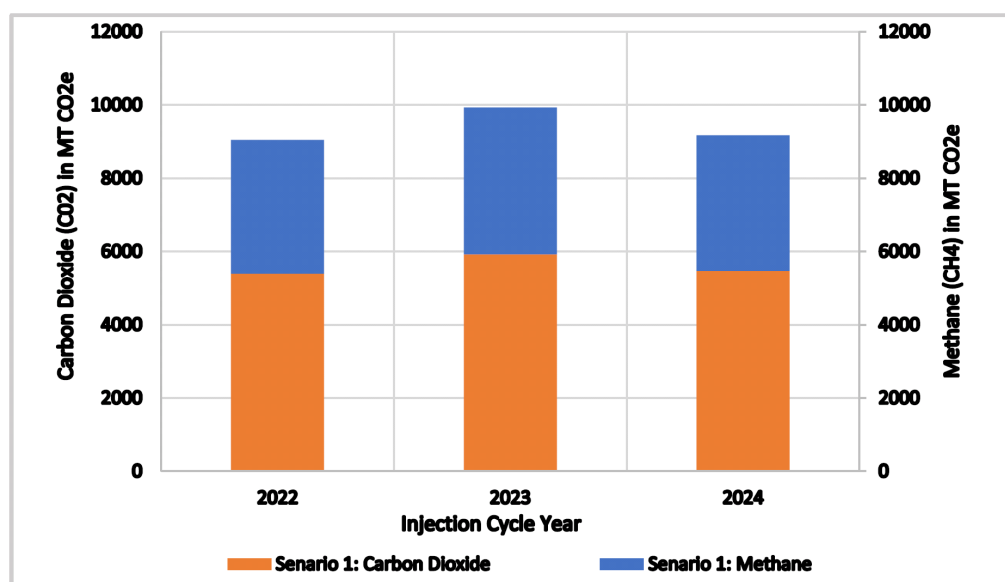


Figure 6—Model Outputs for CO₂ and CH₄ Generation Expressed in Metric Ton of CO₂ equivalent (MT CO₂e) Under Scenarios 1

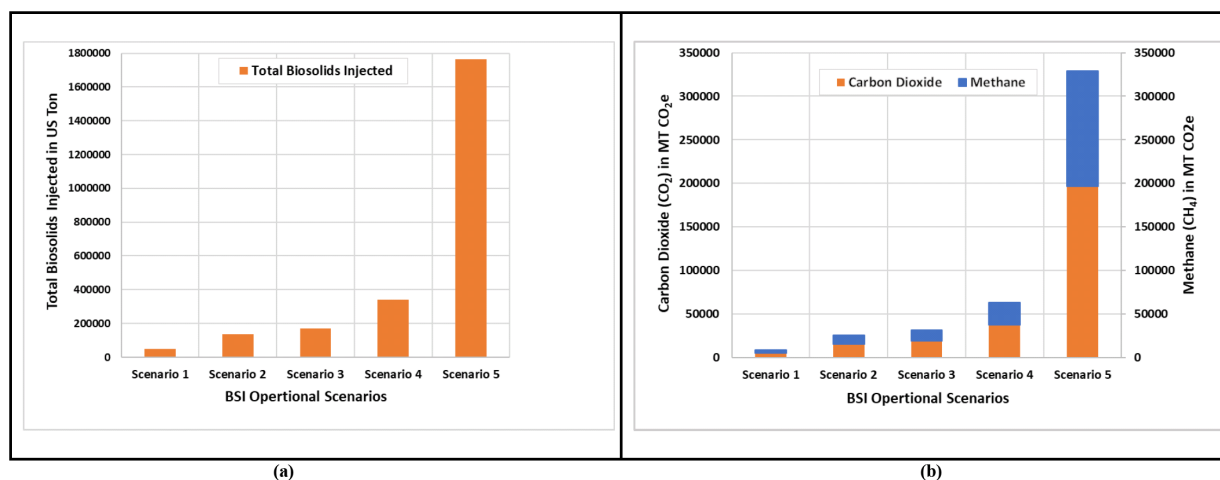


Figure 7—Model Outputs Under Scenario 1 and compared with sensitivity analysis Scenarios 2, 3, 4, and 5 (a) Total Biosolids Injected in US Ton (b) CO₂ and CH₄ Generation (MT CO₂e)

Model Results: GHG Emissions vs. Carbon Dioxide Removal (CDR) in BSI Scenarios

Figures 8 to 9 show TIRE's baseline scenario (scenario 1), detailing total Biosolids injection volume, GHG offset (MT CO₂e over 20- and 100-year GWPs), and carbon offset (MT CO₂e-20 years GWPs), covering the period from 2022 to 2024. Figure 10 and 11 offer a comparative analysis of all BSI TIRE operation scenarios, evaluating GHG offsets and the potential for carbon removal (CDR), providing a comprehensive overview of the TIRE's environmental impact under various operational configurations. The model results reveal a strong correlation between biosolid injection volume and GHG offset, with the highest offsets observed in scenario 5. This highlights TIRE's large-scale processing capacity, maximizing environmental benefits through increased GHG sequestration and delivering significant economic advantages via higher carbon credits associated with substantial CDR. Table 8 summarizes all model output scenarios for CO₂ and CH₄ generation, GHG offsets, and carbon removal (CDR) across all BSI operational scenarios.

In conclusion, the model emphasizes the critical role of scalability within a robust carbon offset framework. Operating at optimal capacity, up to 100%, and supporting large-scale BSI operations, are crucial for managing substantial Biosolids volumes and maximizing GHG sequestration. This approach offers significant promise for both economic viability and environmental benefits, demonstrating that high-volume injection strategies can deliver considerable societal advantages across regions.

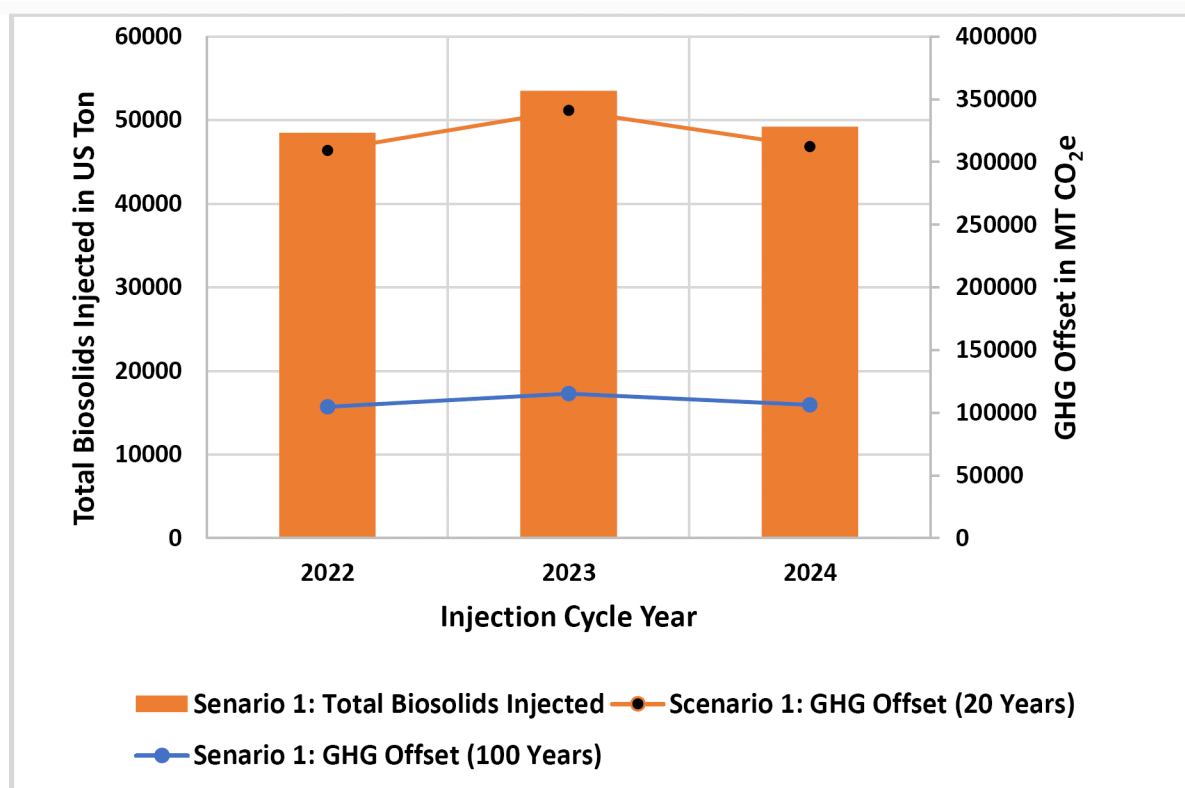


Figure 8—Model Outputs for GHG Offset Expressed in Metric Ton of CO₂ equivalent (MT CO₂e 20 and 100 Years) Under Scenarios 1

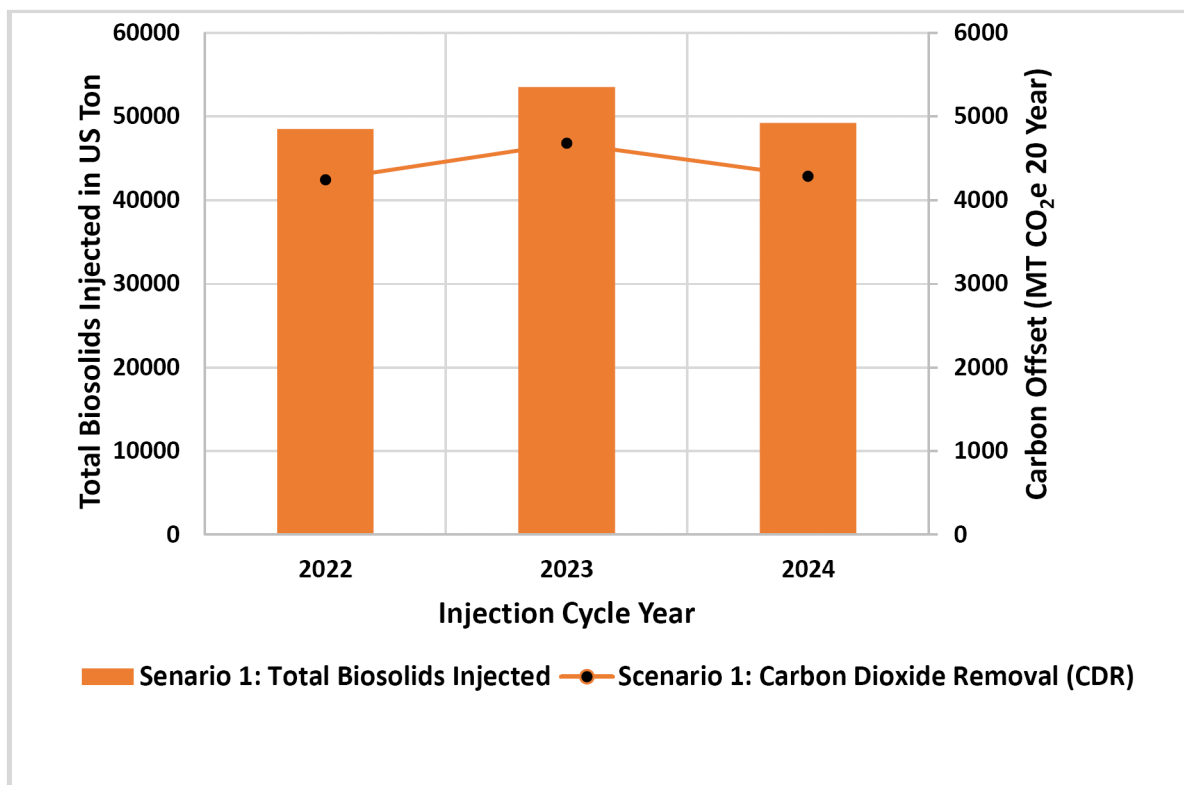


Figure 9—Model Outputs for Carbon Offset (CDR) Expressed in Metric Ton of CO₂e 20 and 100 Years Under Scenarios 1

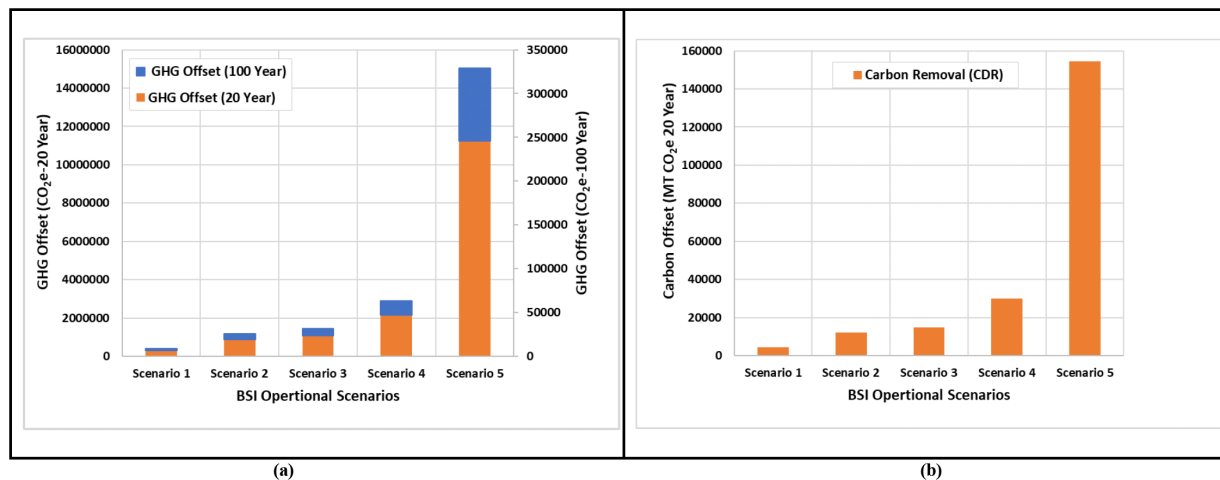


Figure 10—Model Outputs for GHG Offset and Carbon Removal (CDR) Under Scenario 1 and compared with sensitivity analysis scenarios 2, 3, 4, and 5

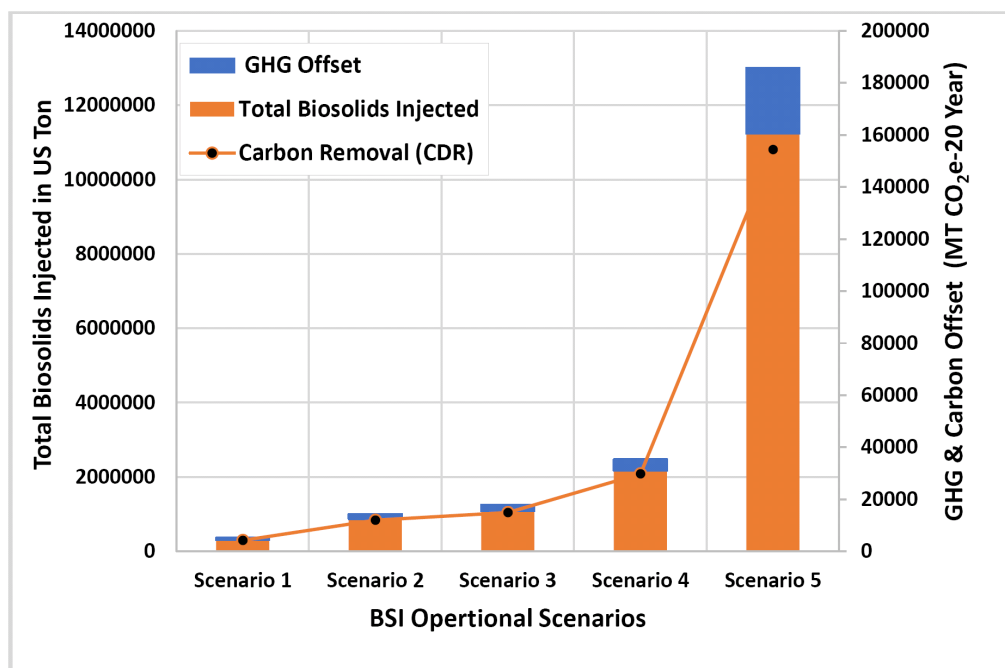


Figure 11—Compression of All Scenarios for GHG Offsets and Carbon Credits

Table 8—Model outputs for CO₂ and CH₄ generation, GHG offsets, and carbon removal (CDR) across all BSI operational scenarios

No.	Scenario 1			Sensitivity Analysis Scenarios			
Year	2022	2023	2024	Scenario 2	Scenario 3	Scenario 4	Scenario 5
Total Biosolids Injection (US-tons)	48,506	53,519	49,247	136,200	170,250	340,500	1,764,698
Total Biosolids Injection (MT)	44,017	48,357	44,654	123,593	154,492	308,984	1,601,359
CO ₂ (MT CO ₂ e)	5,393	5,925	5,471	15,144	18,929	37,859	196,210
CH ₄ (MT CO ₂ e)	3,650	4,010	3,703	10,250	12,813	25,625	132,807
GHG Offset (MT CO ₂ e-20 Year)	309,315	341,329	312,417	867,882	1,084,756	2,169,126	11,240,251
GHG Offset (MT CO ₂ e-100 Year)	104,915	115,276	106,424	293,879	367,252	734,119	3,803,087
Carbon Offset (CDR) (MT CO ₂ e-20 Year)	4,245	4,684	4,287	12,031	29,800	29,800	154,442

Positive Economic Impact (i.e., improved Return on Investment, ROI)

$$\text{ROI} = (\text{Net Profit} / \text{Cost of Investment}) * 100$$

The economic impact of BSI operations is substantial. The low capital and operational costs of this technology (see Figure 2) make it a competitive (i.e., relatively higher ROI) alternative to traditional Biosolids management methods such as drying, incineration, or landfilling. For example, in Houston, a city producing approximately 50,000 dry US tons of Biosolids annually, a single injection well could handle nearly 490,000 dry tons over its operational life. Implementing multiple wells within the city could result in

cumulative savings exceeding \$100 million over 25 years, accounting for costs related to replacing existing drying systems after their operational lifespan (Panchal et al., 2020a and 2020b).

Furthermore, the GHG sequestration achieved through BSI can generate substantial revenue via carbon markets. In California, where carbon offsets are tradable under cap-and-trade systems (Temple, 2019), the GHGs sequestered can be converted into carbon credits. For instance, over the lifespan of a well, the total sequestered GHGs, approximately 180 million MT CO₂e, could translate into over \$180 million in carbon credits, assuming a carbon price ranging from \$1-20 per ton of CO₂e. These credits can be sold to industries seeking to offset their emissions, creating an additional revenue stream. However, fluctuations in carbon prices introduce variability and uncertainty into the potential revenue (Nuccitelli, 2018; Ho et al., 2008).

Impact of Carbon Price Volatility

Global carbon pricing mechanisms leverage two primary approaches: regulated compliance markets and carbon taxation. Compliance markets, exemplified by the European Union Emissions Trading System (EU ETS) and Carbon Border Adjustment Mechanism (CBAM), operate via emissions permit trading, with EU allowances averaging €73/tCO₂e as of August, 2025. Carbon taxes, conversely, impose a fixed per-ton fee, exhibiting wide jurisdictional variance (€1 to > €125).

The voluntary carbon market (VCM) facilitates unregulated trading of verified emission reduction credits, with prices varying significantly, averaging approximately \$6.34 per ton in 2024 but exceeding \$160 per ton for specialized removals like Biosolids and biochar. Emerging mechanisms, including Article 6 trading and results-based climate finance, aim to integrate international carbon mitigation efforts.

In summary, CBAM acts as an avoidable cost, the VCM as a potential revenue stream, and domestic mechanisms as supplementary fiscal tools, depending on policy scope. However, fluctuations in carbon prices introduce considerable variability and uncertainty into potential revenues across regions, affecting investment and implementation strategies.

Positive Environmental Impact

Implementing the BSI technique in TIRE's Facilities in LA City offers significant advantages in reducing greenhouse gas (GHG) emissions. During biogas generation, organic matter in Biosolids decomposes, releasing gases mainly CO₂, CH₄, and N₂O. These gases generation are directly linked to various treatment stages and the volume of organic matter processed (Panchal et al., 2021). Under high-pressure, high-temperature, oxygen-free conditions that inhibit aerobic bacteria, anaerobic microorganisms thrive and decompose organic matter within Biosolids. Laboratory studies of University of California, Los Angeles (UCLA) samples confirm a diverse microbial community, including thermophilic and mesophilic bacteria, with over 67% of the archaeal community belonging to the methanogenic Euryarchaeota phylum (Chen et al., 2005). This breakdown reduces chemical and biological oxygen demand, yielding biogas primarily composed of methane (CH₄) and carbon dioxide (CO₂). Experimental data, supported by lab analyses, indicate that roughly 1 mass unit of Biosolids decomposes into 0.5 units of CO₂ and 0.425 units of CH₄ by mass. Given methane's greater global warming potential, GWP (1 MT CH₄ = 25 MT CO₂e over a 20-year horizon (Gohar and Shine, 2007)), the TIRE facility has cumulatively sequestered over 792,300 US tons of CO₂e to date through BSI operations.

By integrating BSI within LA County, the organic content used in slurry preparation (i.e., Bioslurry) can be efficiently collected and accumulated at each stage. This approach streamlines Biosolids management and significantly reduces GHG emissions by minimizing organic material (i.e., Biogas) decomposition and methane release into the atmosphere. Furthermore, since the Biosolids are injected underground and not transported to land application sites, the associated truck miles and consequently, transport-related air pollution and emissions are substantially reduced.

To accurately quantify the environmental benefits of the BSI process, a dedicated model estimates annual GHG savings. This model converts all relevant emissions, including those from sludge treatment, decomposition, transportation, and electricity consumption, into CO₂ equivalents (CO₂e) and aggregates them to provide a holistic assessment.

Based on data from the TIRE facility under the current operational scenario (scenario 1), the average annual Biosolids received from LA County is approximately 50,000 US short tons. This process helps prevent the release of approximately 341,329 metric tons of CO₂e emissions over a 20-year period, contributing significantly to greenhouse gas reduction efforts.

Expanding operations to fully utilize 100% of TIRE's capacity (scenario 5) would enable the processing of approximately 1,764,698 US short tons of Biosolids. This scale-up could achieve a total GHG offset of approximately 11,240,251 MT CO₂e over the same period. Furthermore, this increased capacity would significantly enhance economic returns by generating higher carbon credits, potentially exceeding those of scenario 1 by more than 30-fold, amounting to roughly 154,442 MT of CO₂e. These findings underscore the substantial environmental and financial benefits of scaling up TIRE's operations.

Beyond climate and economic benefits, BSI enhances resilience against natural disasters. Coastal Sewage Sludge Treatment Facility (SSTF), often located in vulnerable low-lying areas, face risks from flooding, hurricanes, and rising sea levels, which can damage infrastructure and cause the release of pollutants (Friedrick and Kretzinger, 2012; Flood and Cahoon, 2011). The BSI process enables rapid response to such events, allowing for the underground injection of the entire biosolid load during forecasted storms, thus effectively preventing flooding-related damage and environmental contamination.

In summary

Decades after recognizing the threat of GHGs, tangible large-scale atmospheric reductions and associated impact mitigation remain a distant prospect. Within the diverse range of climate mitigation strategies, CDR and storage techniques, such as BSI, have become promising and readily implementable options for offsetting the detrimental effects of anthropogenic (human-caused) CO₂ emissions. Since commencing operations in 2008, the TIRE facility has injected over 972,300 US-tons of Biosolids into deep geological formations. This operation offsets approximately 300,000 metric tons of CO₂-equivalent emissions annually, yielding a cumulative mitigation of about 5 million metric tons of CO₂-equivalent over its 17-year operational history. This analysis underscores the scalability of subsurface injection to meet metropolitan carbon removal goals, emphasizing the importance of monitoring formation integrity, injectivity, and plume migration to ensure containment and optimize storage efficiency.

Energy Generation and Societal Advantages

Beyond the environmental and economic advantages, BSI offers significant social benefits. By injecting Biosolids within Sewage Sludge Treatment Facility (SSTF) boundaries, the technique eliminates the noise and air pollution associated with transporting Biosolids to remote land application sites. As urbanization brings communities closer to sewage sludge treatment facility, odor from biosolid dewatering has become a growing concern. For example, in 2013, the Texas Commission on Environmental Quality received a petition from Ellis County to prohibit sludge land application due to odor issues (Panchal et al., 2020a; Hirst, 2014). Thus, implementing BSI technique addresses this directly by eliminating both the dewatering stage and land application, thereby resolving the odor problem. Furthermore, the anaerobic decomposition of injected Biosolids generates CH₄ underground. This methane, if recovered, could provide a valuable energy source for on-site electricity generation, heating, or even as a cooking fuel, contributing to energy independence and resource recovery.

Environmental Risk (Risk and Risk Assessment)

At large scale, BSI operations may present certain potential environmental challenges. First, as with any underground carbon storage approach, there's a risk of leakage and subsequent re-emission of sequestered carbon into the atmosphere. This would require anaerobic metabolism by subsurface bacteria, a process with an uncertain rate. Even if anaerobic digestion occurs, trapping mechanisms within the reservoir could retain resulting CO₂ and CH₄; however, fracturing could damage structural trapping, potentially compromising long-term containment (Kordzi, 1998).

Second, the fracturing and injection process itself could have environmental effects, distinct from those of conventional fracking. Vertical BSI wells generally reducing the likelihood of upward injectate migration. While fluid injection can activate faults and induce seismicity, this is typically associated with large-volume water injection (> 300,000 bbls per month) near the basement rock. BSI (Surface Fracturing Injection) can induce surface uplift and deformation, which is a concern in shallow formations (less than 800 meters) and requires monitoring (Weingarten et al., 2015; Bruno et al., 1995; Hincks et al., 2018; Sipple-Srinivasan et al., 1997; Germanovich and Murdoch, 2010). Therefore, for BSI, injection is typically conducted below 900 meters to mitigate surface deformation risks.

Third, scaling up TIRE's operations necessitates substantial biosolid quantities. If this biosolid were cultivated on prime agricultural land, required extensive transport, or relied on significant fertilizer or freshwater inputs, negative impacts on food supplies and other environmental systems could arise. However, BSI's ability to utilize diverse feedstocks, including Biosolids, has the potential to mitigate these systemic environmental concerns. Table 9 summarizes risk categories and corresponding mitigation strategies for subsurface biosolid management options within BSI technology.

Table 9—Biosolids Management Option: Biosolid Slurry Injection (BSI) (Abou-Sayed et al., 2002; Abou-Sayed and Guo, 2002; Abou-Sayed and Guo, 2001; Panchal et al., 2021; Panchal et al., 2020a&b; Panchal et al., 2018)

Risk Category	Specific Hazard Description	Primary Controls / Mitigations
Environmental	Migration of Biosolids or gases into underground sources of drinking water (USDW) poses a potential risk of water contamination.	Class I/II/VI-style designs ensure safety and containment, supported by high-integrity casing and quality cement to prevent leaks and maintain well integrity over time. Regular Mechanical Integrity Testing (MIT), coupled with continuous monitoring and formation stability management, safeguards long-term well performance
Health & Safety	H ₂ S may generated during Biosolids decomposition under anaerobic conditions, releasing a toxic and flammable gas, posing significant health and safety risks.	Ongoing monitoring of hazardous gases through real-time gas detection systems;
Geotechnical/ Seismic	Induced seismicity due to increased subsurface pressure.	Step-rate tests (SRT) determine safe pressure thresholds, while real-time seismic monitoring identifies seismic activity during the BSI
Social/Community	Groundwater (USDW) risk perception centres on concerns about Biosolids leaching into drinking water sources, while induced seismicity relates to seismic events may triggered by injection activities.	Public dashboards enable real-time transparency, while grievance mechanisms facilitate prompt response to community concerns.
Regulatory/Legal	Permitting complexity and public acceptance require regulatory approval and community engagement, while long-term stewardship ensures ongoing safety and environmental protection post-project.	Early engagement with regulators; securing financial assurance; and implementing transparent monitoring plans.
Technical/ Operational	Formation incompatibility can lead to clogging, causing injection problems, and souring due to adverse fluid-rock chemical interactions	Geochemical testing evaluates fluid-formation reactions; pre-treatment reduces biological activity; stimulation enhances injectivity; combined chemical-geochemical simulations improve formation response understanding.

Conclusions and Future Directions

The BSI process demonstrates a viable, scalable solution for sustainable Biosolids management, GHG mitigation, and climate change adaptation. It confirms that slurry injection offers a safe, cost-effective, and environmentally sustainable alternative to conventional Biosolids disposal management options such as land application, landfilling, and incineration. Operational data from the TIRE facility exemplifies its capacity to sequester substantial volumes of CO₂ and reduce atmospheric GHGs. Over its 17-year history, the facility has injected approximately 972,300 US tons of Biosolids, offsetting roughly 300,000 metric tons of CO₂e annually and amounting to a total reduction of about 5 million metric tons over the operational lifespan.

Economically, BSI offers significant advantages. Its low capital and operational costs yield a high return on investment, outperforming existing NETs options, including DAC, BE-CCS, CCUS, Biochar, and Enhanced, Weathering. Moreover, a single-well configuration processing 150 US-tons per day can generate substantial annual carbon credits, estimated at over 85,000 (MT of CO₂e-20 Year), although revenue potential remains sensitive to market price fluctuations. Scaling to the full capacity (100%) of the TIRE facility, with a seven-well configuration capable of processing 7,774 US-tons daily, could achieve annual GHG offsets exceeding 11 million (MT of CO₂e-20 Year) and a carbon credits or CDR capacity of approximately 160,565 (MT of CO₂e-20 Year). By reducing transportation logistics and land application requirements, BSI also minimizes emissions and community impacts such as odors and noise pollution, further enhancing its sustainability profile.

From an environmental perspective, BSI operation provides a safe, long-term sequestration pathway within geologically suitable formations, minimizing risks of groundwater contamination and surface pollution. Well design and rigorous monitoring are essential to prevent upward gas migration and induced seismicity, particularly in regions with favourable geology. Additionally, BSI enhances disaster resilience in coastal areas by enabling rapid underground injection during forecasted storms, thereby mitigating flood risk and environmental damage.

Social benefits for regional communities include enhanced urban air quality, pollution control, odor reduction, and reduced exposure to Biosolids during handling and transportation. In areas where land application or landfilling pose environmental or social challenges, BSI provides a sustainable and socially acceptable alternative aligned with urban growth and public concerns.

Nonetheless, potential risks such as subsurface gas leakage, induced seismicity, and sourcing large biosolid volumes warrant careful management. Ensuring containment integrity and establishing comprehensive risk mitigation strategies are critical to address these challenges.

In summary, when implemented with site-specific assessments, proper regulation, and ongoing monitoring, BSI presents a safe, cost-effective, and environmentally beneficial approach for large-scale Biosolids management. Its capacity to generate negative emissions positions it as a key component in regional and global climate mitigation efforts. Future research should focus on optimizing methane recovery, long-term sequestration validation, and risk management to maximize the BSI technology's sustainability and societal acceptance.

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NOMENCLATURE

Symbols

CO	Carbon Monoxide
CO ₂	Carbon Dioxide
CO ₂ e	Carbon Dioxide Equivalent; CO ₂ e value of 1
CH ₄	Methane
HCl	Hydrogen Chloride
N ₂ O	Nitrous Oxide
NO ₂	Nitrogen Dioxide
PM	Particulate Matter, Airborne Particles such as Dust (not a Chemical Formula)
SO ₂	Sulphur Dioxide

Subscripts/Superscripts

G	Gas
Gt	Giga-ton, equal to one billion (10 ⁹) tons
g	Gram
gal	Gallon
kg	Kilogram
O	Oil
m	Meter
mpgal	Mile per gallon
t	Ton
W	Water
ya	Year

Abbreviations

APC	Air Pollution Control
BE-CCS	Bioenergy with Carbon Capture and Storage
BSI	Biosolid Slurry Injection
CA DOF	California Department of Finance
CAPEX	Capital Expenditures
CCUS	Carbon Capture, Utilization, and Storage
CHP	Combined Heat and Power System
EPA	Environmental Protection Agency
EMS	Emissions Monitoring System
GHG	Greenhouse Gas
GWP	Global Warming Potential
GWP-20	Use a 20-year Time Horizon
GWP-100	Use a 100-year Time Horizon
HWRA	Hyperion Water Reclamation Plant
IPCC	Intergovernmental Panel on Climate Change
K	Bioenergy with Carbon Capture and Storage
LA	Los Angeles
LEL	Lower Explosive Limit
LFG	landfill gas capture
MT	Metric Ton, which is equal to 1,000 kilograms or 2,204.6 pounds

NET	Negative Emissions Technologies Direct Air Capture
O&G	Oil and Gas
OPEX	Operational Expenditures
PFAS	Polyfluoroalkyl substances
PPIC	Public Policy Institute of California
ROI	Return of Investment
SI	Slurry Injection
SCF	Standard Cubic Foot
SSTF	Sewage Sludge Treatment Facility
UCLA	University of California, Los Angeles
UIC	Underground Injection Control
US	United States
US EPA	United States Environmental protection Agency
UWWTD	Union's Urban Waste Water Treatment Directive

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